Paper Review • arXiv 2610.40636

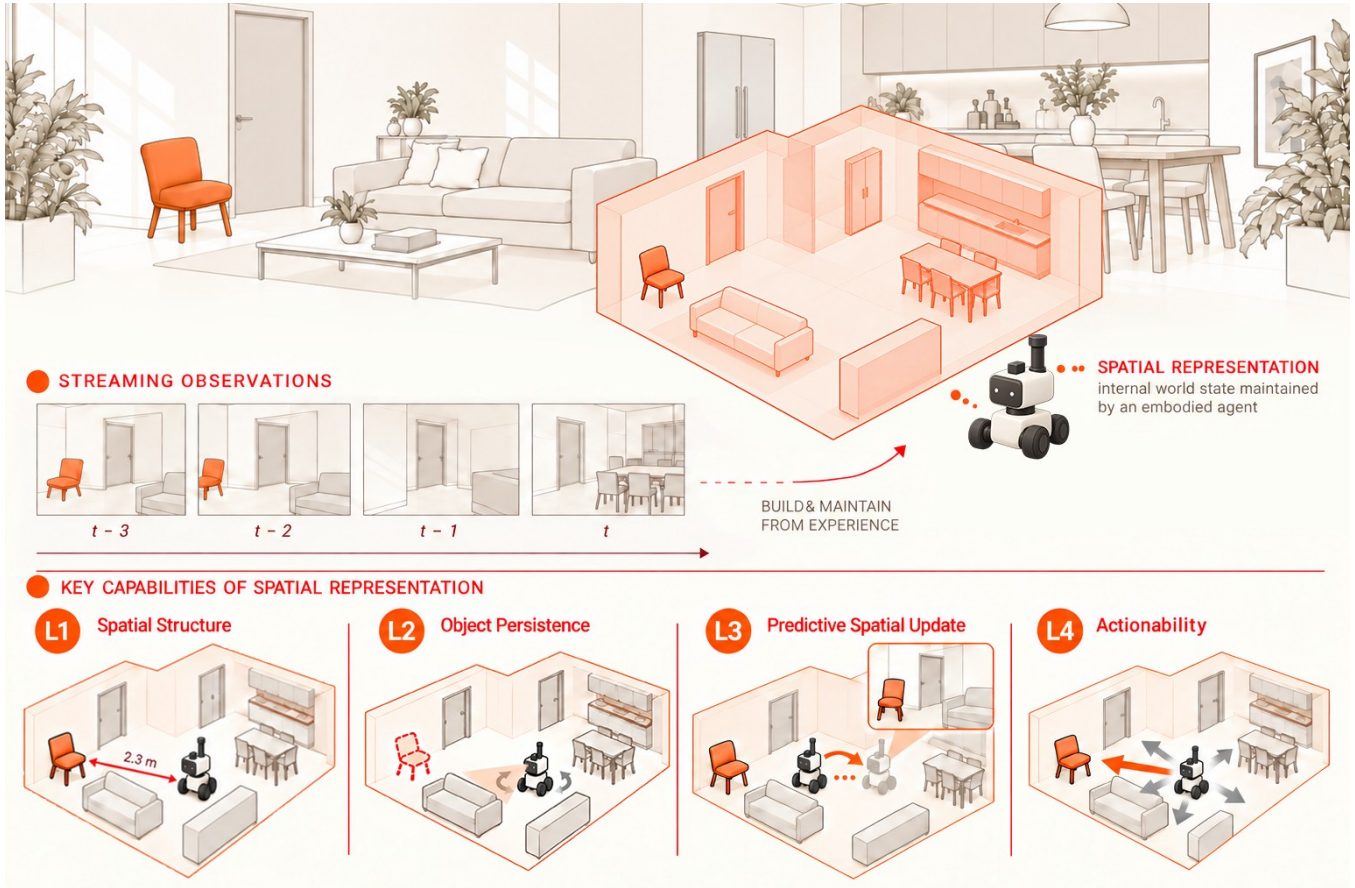
Probing Spatial State Representations in Visual Foundation Models

Do visual foundation models hold a spatial state that can remember, predict and act?

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Source: Figure 1 (streaming egocentric observations and four levels of spatial capability)

Key Findings

In short: decodable ≠ used ≠ useful

Frozen-feature probes test four levels of spatial capability in 10 visual foundation models on streaming egocentric ScanNet++ video: no model leads overall, and decodability, input use and functional benefit to the goal are dissociated.

Decodability

Decodability

VideoMAE localization error is 36.5°, far better than DINOv2's 65.4°, despite DINOv2's stronger depth/instance/view readouts

Input use

Input use

VideoMAE localizes best yet shows no benefit from history (+2.1°, CI spans 0); for V-JEPA2, copying the last frame beats learned prediction

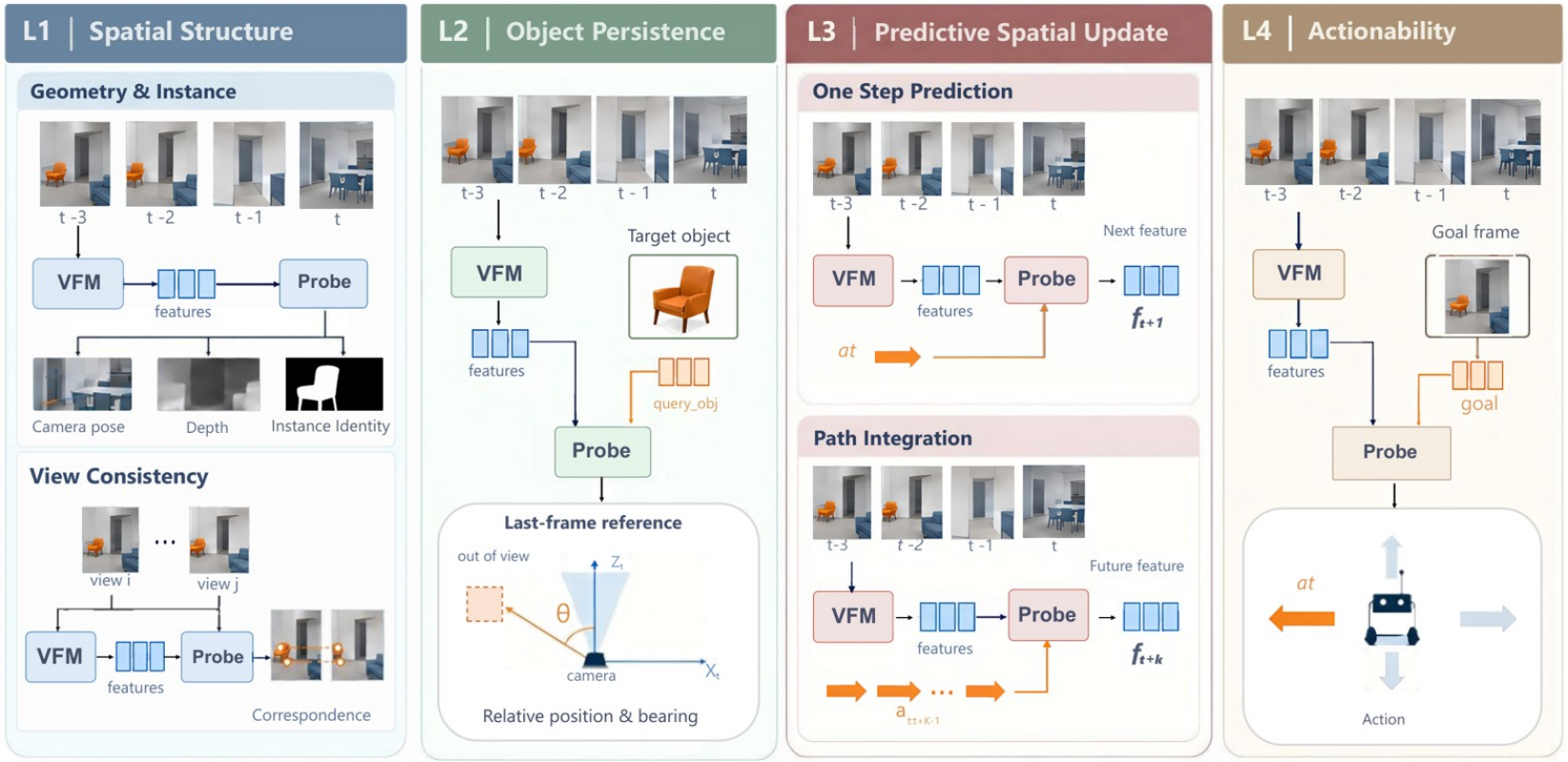
Functional utility

Functional utility

Swapping goals deflects predicted actions by 30–46°, but the correct goal changes progress by only −2.3 to +1.2pp; all CIs include 0

Problem Formulation

Decomposing "spatial state" into four progressive functional levels



L1 Spatial structure

Reads only the visible scene

L2 Object permanence

+ Retains earlier views

L3 Predictive update

+ Predicts unseen frames

L4 Actionability

+ Links state to goals and actions

Experimental Setup

10 frozen models × causal streams × shallow probes

Image encoders	DINOv2-L	
Video encoders	V-JEPA2 ViT-L	VideoMAE-L
3D encoders	Fast3R ViT-L	
Multimodal	Qwen2.5-VL-7B	
Video generators	Wan2.1-1.3B	CogVideoX-5B
	MiniMax-H3	LingBot-video
World models	AetherV1	

$p \in \{8,12,16,24\}$

Causal prefix lengths; each prefix encoded independently

69 scenes

Validation set, disjoint from training; 1,000 scene-level paired bootstraps

276 windows/prefixes

Main table; 828-candidate direct/path pool

7 measures

6 core diagnostics (geometry = depth + pose); +2 L4 settings

All backbones frozen; only shallow probes trained. Main table uses a fixed designated layer, averaged over prefixes, with no post-hoc best-layer selection.

No model leads on every spatial task

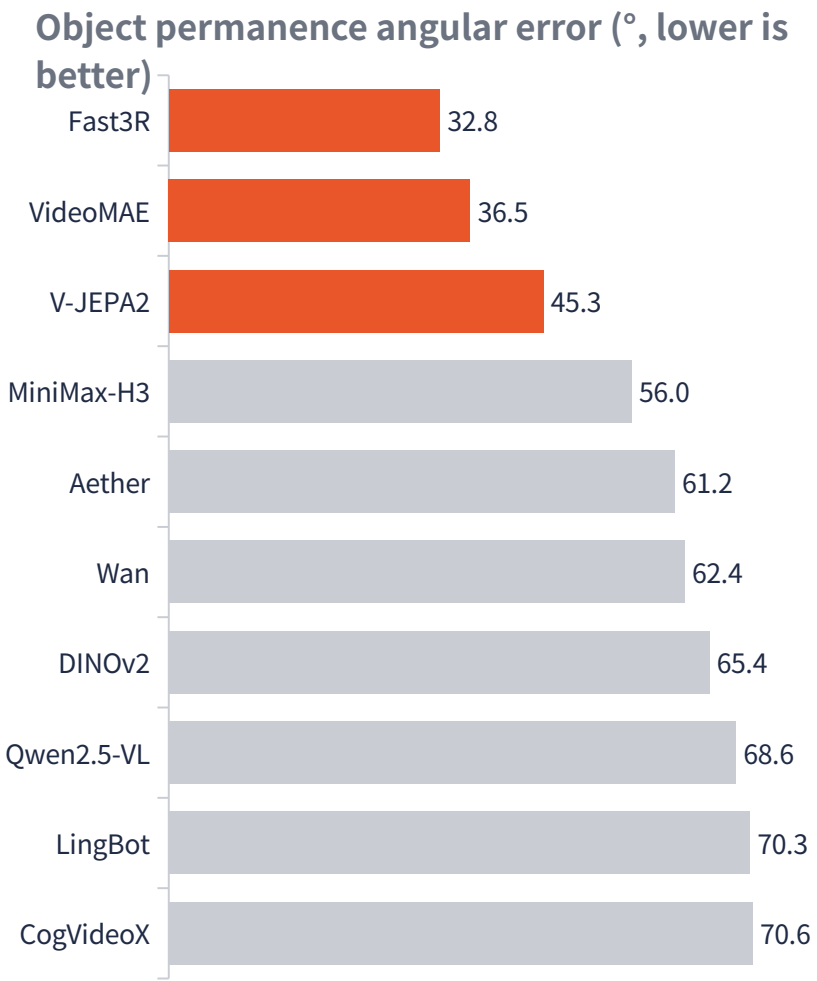
Model (layer)	Depth loss ↓	Pose AUC@30 ↑	Instance loss ↓	View consist. ↑	Permanence err. ° ↓	Direct R@1 ↑	Path R@1 ↑
DINOv2-L (0)	**0.1422**	0.0412	**1.3828**	0.9557	65.38	**0.2373**	0.0900
Fast3R (24)	**0.1386**	**0.3870**	1.6072	**0.9808**	**32.79**	0.0133	0.0190
VideoMAE-L (23)	0.2968	0.2026	4.1571	0.9118	36.45	0.0531	0.0685
V-JEPA2-L (23)	0.2708	0.1854	2.9211	0.9256	45.31	0.0625	**0.1588**
Qwen2.5-VL (−1)	0.2244	0.0269	2.9476	0.9385	68.59	0.1712	0.0903
Wan (20)	0.3609	0.0650	4.9301	0.8215	62.38	0.0260	0.0405
CogVideoX (20)	0.4340	0.0276	6.6992	0.8001	70.62	0.0088	0.0160
Aether (1)	0.5270	0.0185	7.1847	0.6928	61.18	0.0024	0.0042
MiniMax-H3 (30)	0.4252	0.0477	6.2265	0.8255	55.97	0.0249 †	0.0117 †
LingBot-video (30)	0.4314	0.0232	5.5875	0.7858	70.30	0.0302	0.0429

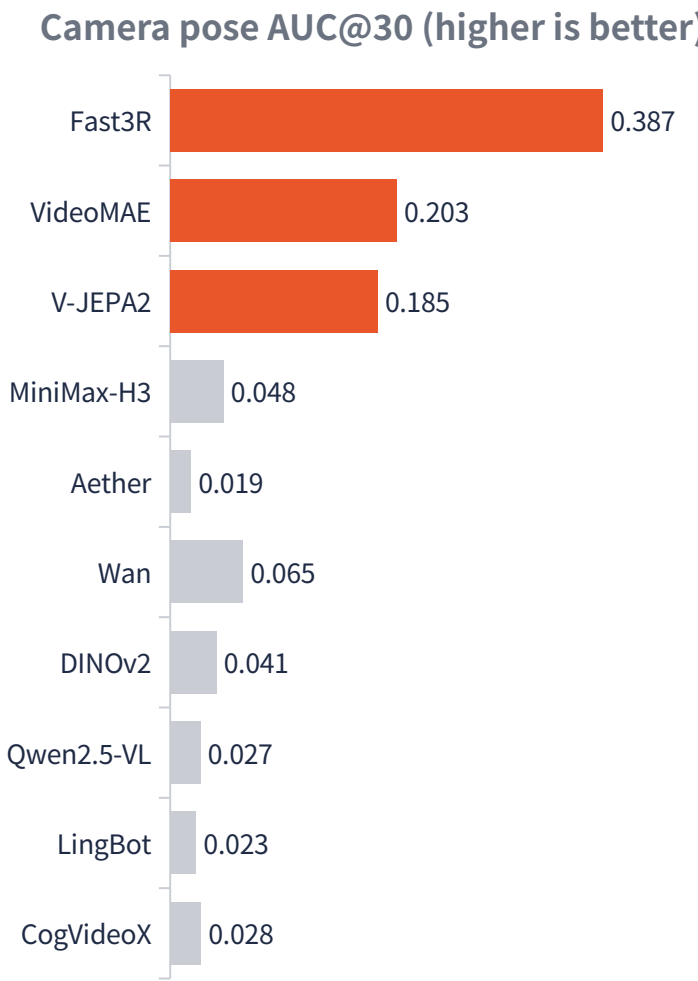
† MiniMax-H3 prediction evaluated on only 171/276 windows; direct prediction uses next-frame targets only; excluded from ranking.

Fast3R tops pose & permanence yet is near last in feature prediction; DINOv2 tops Direct, V-JEPA2 Path integration.

L2 ← L1

Object permanence tracks pose, not scene, readout





ρ = 0.76

Permanence vs camera pose (10 models)

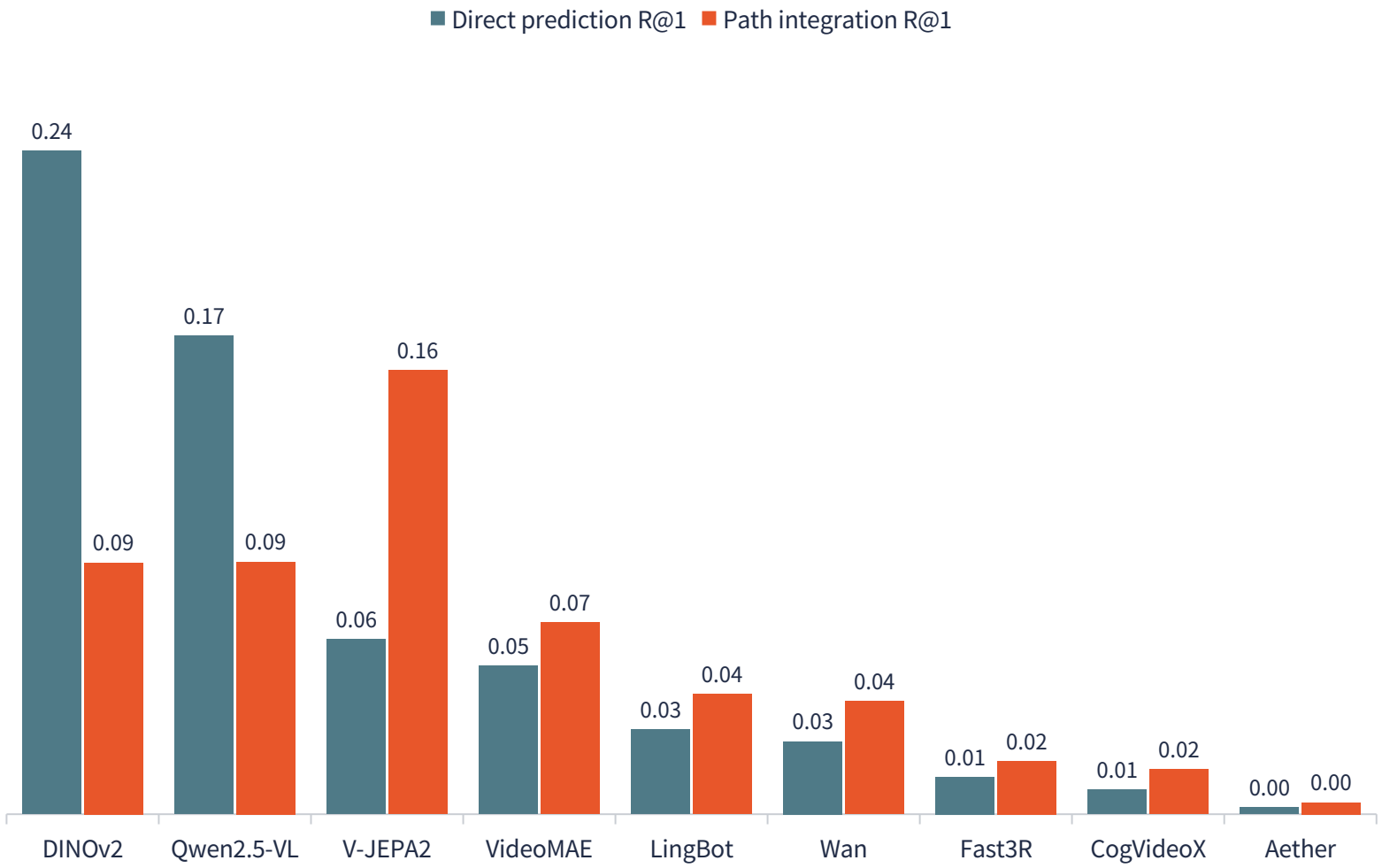
ρ ⩽ 0.43

Permanence vs depth / instance / view

Locating out-of-view objects requires relating the current camera to earlier views; in four multi-layer backbones, permanence and geometry peak at the same layer.

L3

Feature-prediction rankings are unrelated to permanence



ρ = 0.93

Model rankings: Direct vs Path

|ρ| < 0.1

Direct/Path vs permanence

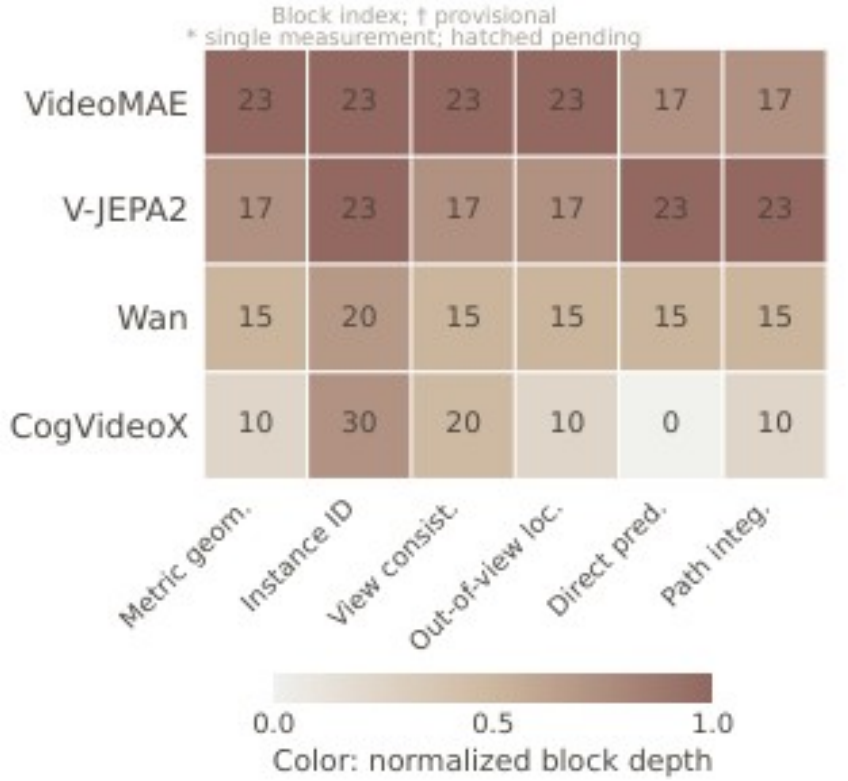
8th / 7th

Fast3R's rank on Direct / Path (best on nearly all L1)

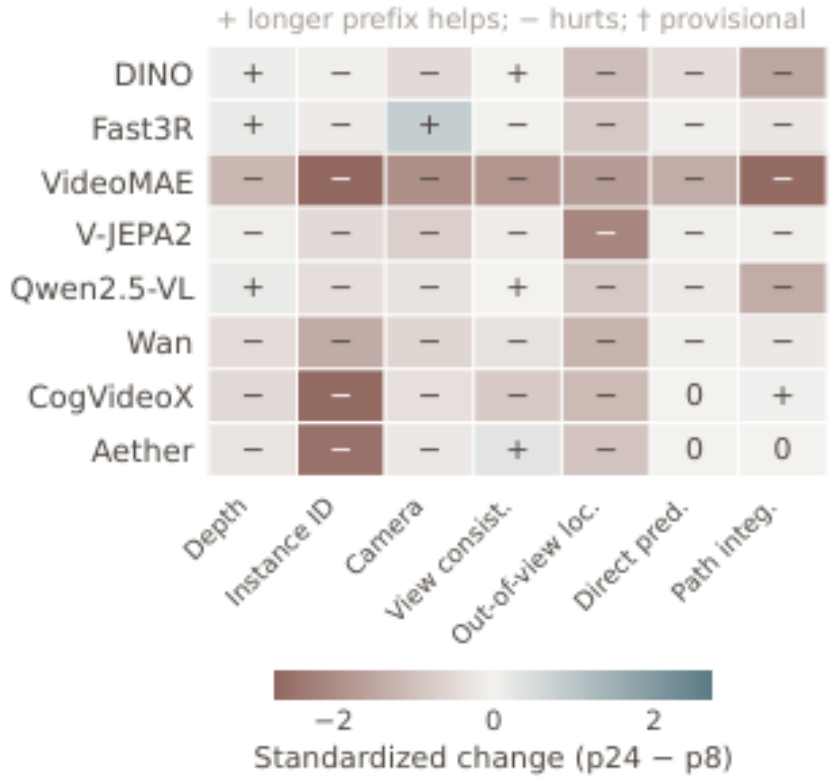
R@1 retrieves within each model's own feature space; only rankings are compared across models.

Setting Dependence

Both extraction layer and prefix length change the conclusions



(a) Best layer per task (color = normalized depth)



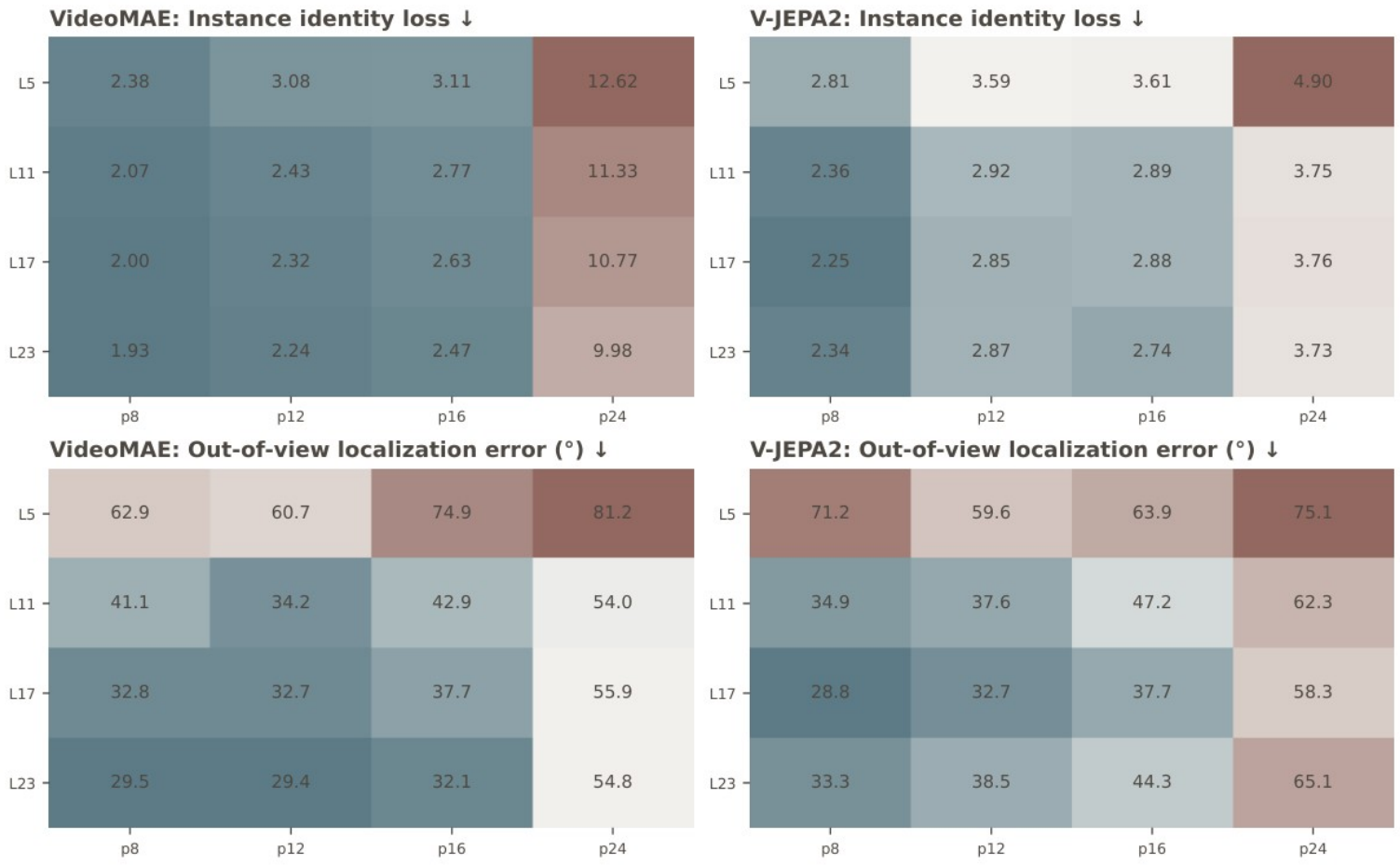
(b) p24 − p8 standardized difference (− = shorter prefix better)

28.8° V-JEPA2 (L17, p8) localization error, below VideoMAE's best of 29.4°

45 / 56 model × metric pairs favor the shorter p8

8 / 8 models' permanence significantly worse at p24

VideoMAE collapses abruptly at p24, while V-JEPA2 degrades gradually



29.6° → 54.8°

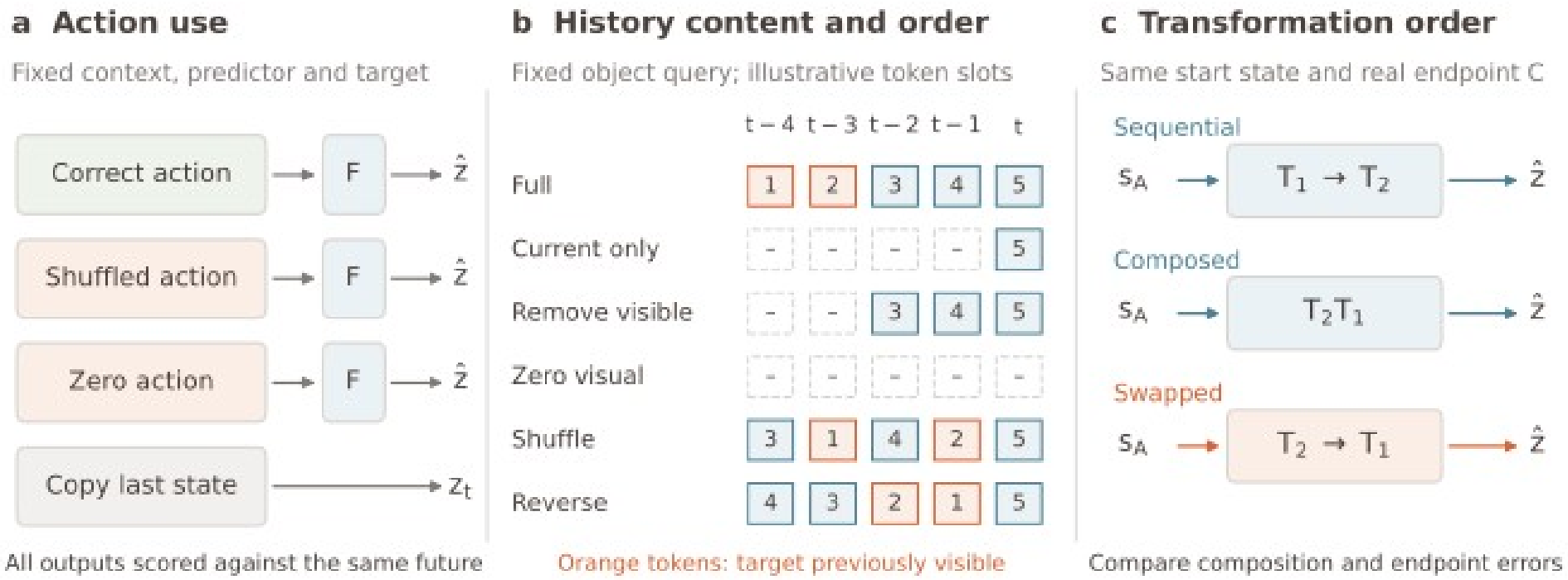
VideoMAE localization error, p8 → p24 (designated layer)

2.47 → 9.98

VideoMAE block-23 instance loss, p16 → p24

Extraction layer and prefix length are part of the evaluation itself, not an interchangeable feature interface.

Matched controls: fix probe and target, vary only the input



a Action use

Correct / shuffled / zeroed actions; or copy last-frame features as a temporal baseline

b History content & order

Current frame only, drop target-visible frames, zero visual tokens, shuffle or reverse frames; object query kept

c Transform order

Sequential T1→T2, composed T2T1, swapped T2→T1, compared against the same true endpoint

Decodable is not used: the best localizer does not use history

Copy-last vs prediction

V-JEPA2 (global retrieval, one step)

19.93% copy

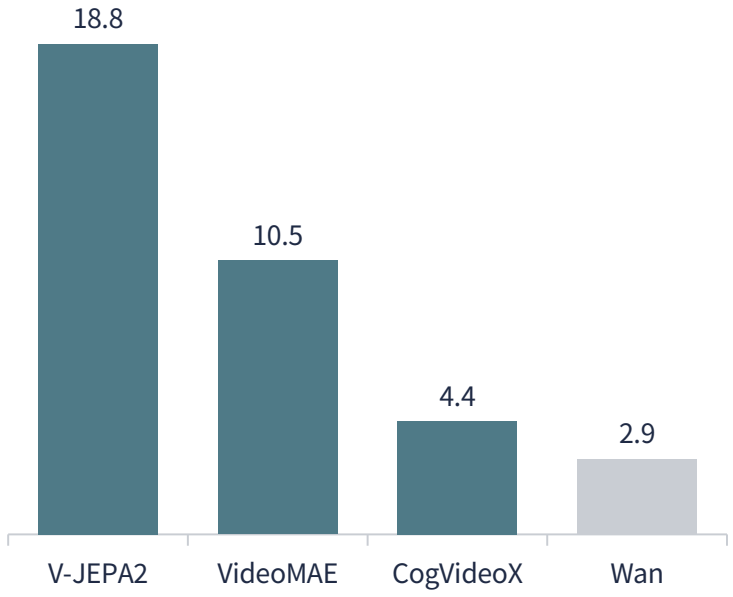
13.04% predict

Wan

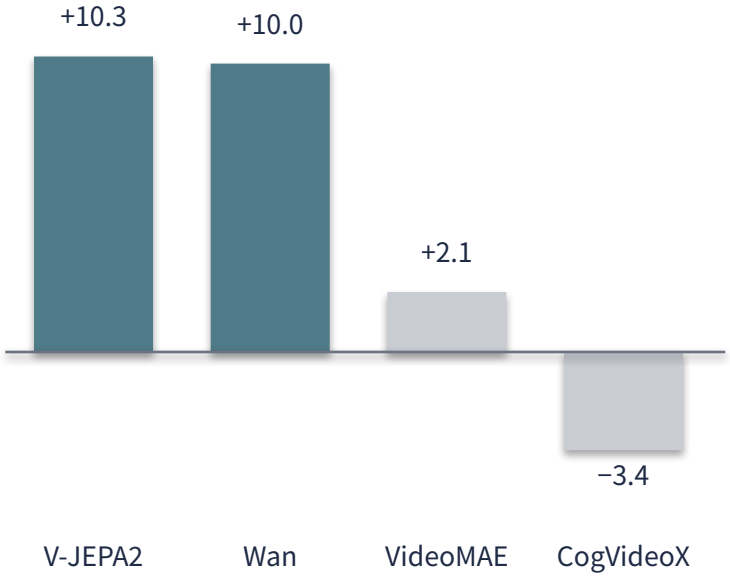
Predict 15.22% > Copy 11.96%

For V-JEPA2, learned prediction is worse than a "do-nothing" copy baseline.

Motion gain (correct − wrong motion, pp)



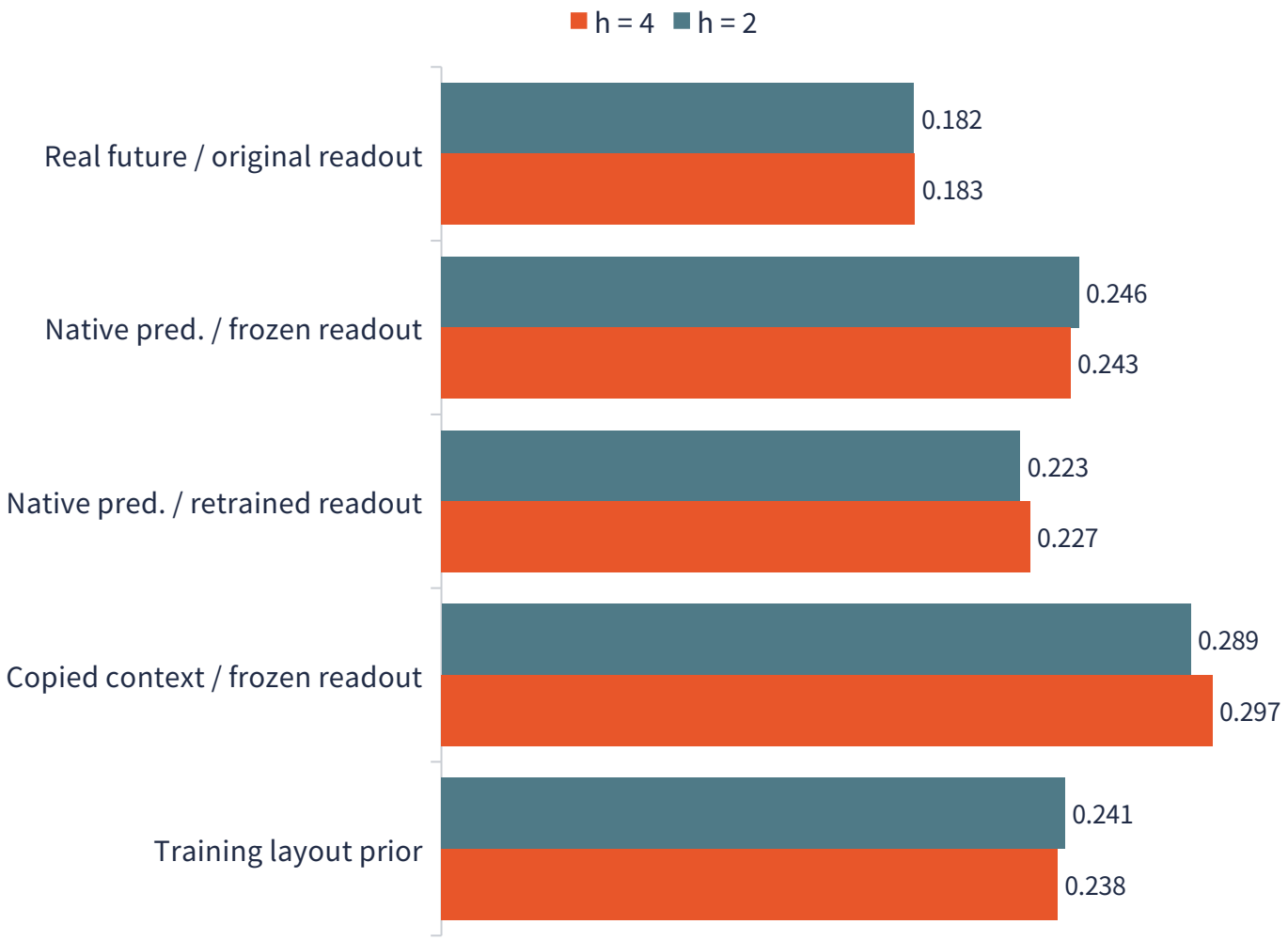
History gain (current frame → full history, °)



Gray = CI spans 0; CogVideoX at −3.41° (not significant)

VideoMAE localizes best (36.5°) yet shows no benefit from history (+2.1°, CI spans 0); shuffling frames has almost no effect: the probe reads history content, not ordered state updates.

Native prediction keeps structure, shifts readout



0.182 → 0.246

h=2: same readout, real features → native predictions

0.223

Retrained readout: beats prior (0.241), still below real features

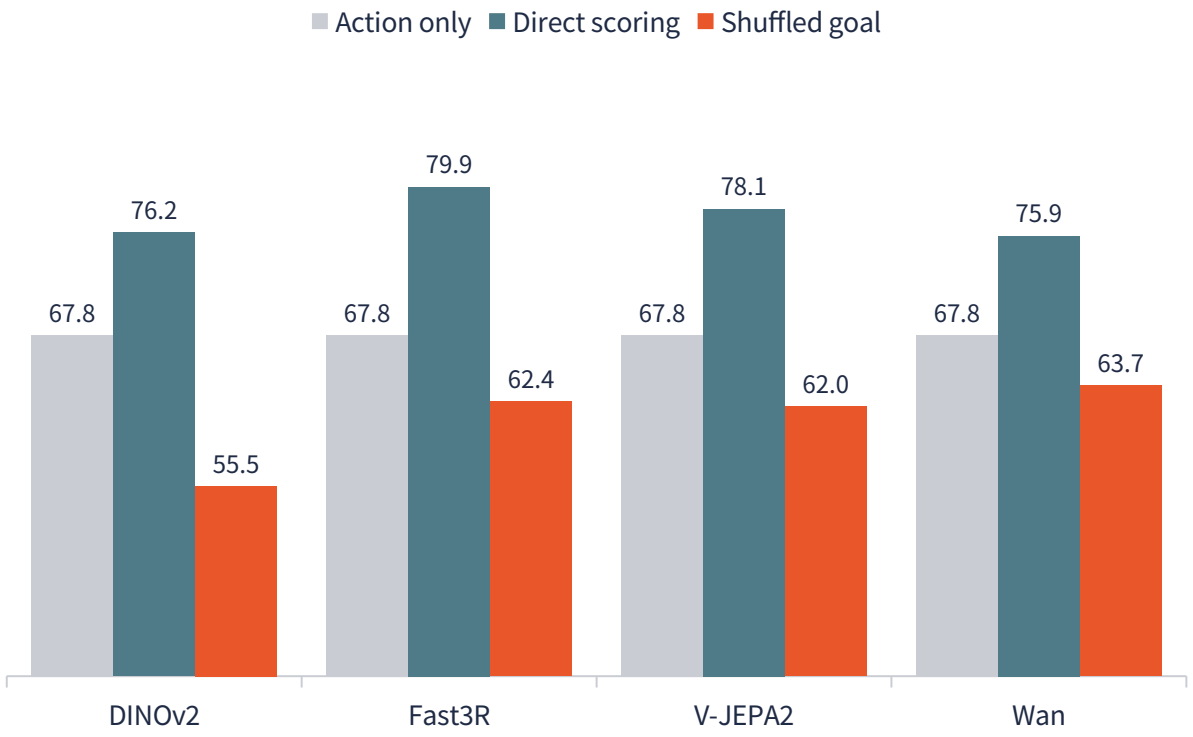
Recoverable ≠ directly readable: prediction is neither lossless spatial propagation nor a total loss of spatial information.

arXiv 2610.40636 • Data: Table 15 / Figure 5(a) (normalized depth MAE, lower is better, mean of 3 seeds)

L4 • Actionability

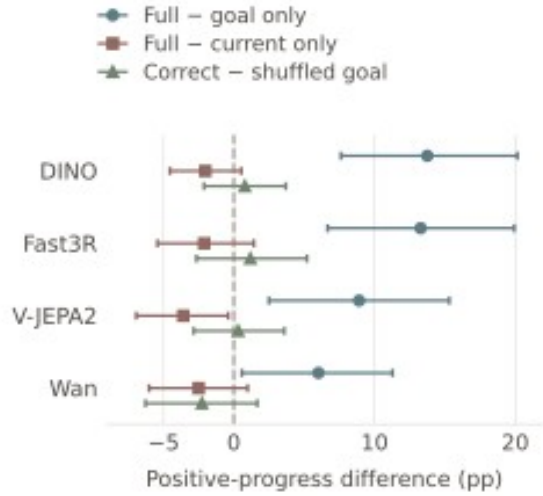
Actions follow goals; correct goals add no progress

Candidate selection accuracy (%, mean of 3 seeds; chance 50%)



Candidate selection does use the goal: shuffling goals lowers accuracy by 12.2–20.7pp.

Continuous-output setting: positive-progress difference (pp)



30–46°

Goal-swap heading shift

−2.3~+1.2

Correct−shuffled progress (pp); CIs incl. 0

FULL beats GOAL-ONLY by 6.0–13.8pp, but no backbone surpasses CURRENT-ONLY; offline diagnostic, not a closed-loop policy.

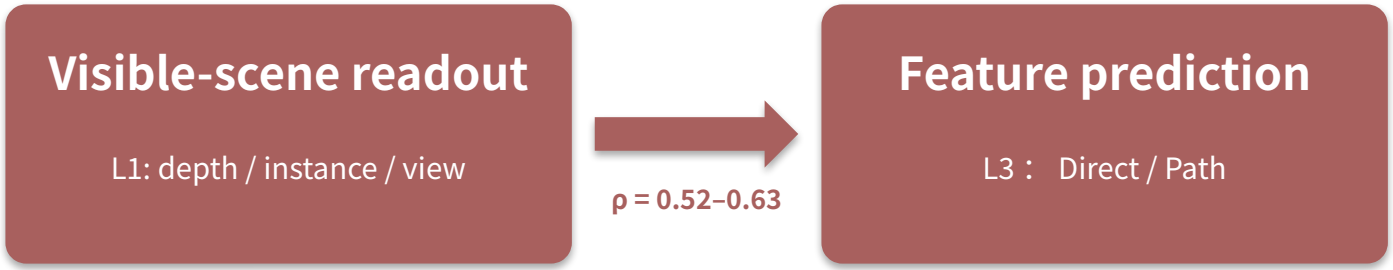
Level Relations

Two "component chains" run through the four levels

Chain A: pose → permanence → action



Chain B: visible scene → feature prediction



L4 ranking: Fast3R > V-JEPA2 > DINOv2 / Wan

Matches permanence (32.8° / 45.3° / 65.4° / 62.4°), unrelated to Direct prediction (DINOv2 first, Fast3R last)

Lower levels are necessary but not sufficient for higher ones: Fast3R is best on nearly all L1, yet ranks 8th / 7th in prediction.

Summary

Takeaway: profile spatial representations, not a single score

Main Findings

- No overall winner: Fast3R leads pose and permanence, DINOv2 direct prediction, V-JEPA2 path integration
- Layer, prefix and generative coordinates all reshuffle rankings; longer prefixes generally hurt permanence
- Decodable ≠ input used ≠ functionally helpful for the goal
- Two component chains: pose → permanence → action; visible scene → feature prediction

Limitations & Caveats

- Prefixes encoded independently, so online recurrent state is untested; prefix length is confounded with endpoint position
- Static indoor scenes only (ScanNet++); Fast3R may have seen some validation scenes
- Conclusions depend on shallow probes and layer / coordinate choices; cross-model R@1 supports ranking only
- L4 covers only 4 models, offline diagnostics; component chains are correlational, not causal, evidence

Evaluating spatial representations means asking not only what can be read out, but whether information persists out of view, updates predictively with motion, and truly serves goal-directed action.